Category Theory for Mathematical Physics — Exercise Sheet 1

Categories: a theory of systems and processes

Variant: Solutions

Conventions as in Lecture 1: in $g \circ f$ the morphism f acts first; BG has one object $*$ with $\mathrm{Hom}(*,*) = G$ and $g \circ h = gh$; $\mathrm{Rep}(G)$ has the finite-dimensional complex representations and all intertwiners. A *homomorphism* of groups preserves products (and hence the identity). The cyclic group $\mathbb{Z}_n$ is written additively. In Exercise 3, $\omega = e^{2\pi i/3}$.

Exercise 1 (Categories, non-examples, and groups). (i) Take as objects the natural numbers $0, 1, 2, \ldots$, with exactly one morphism $m \to n$ when $m \leq n$ and none otherwise. Show that this is a category, and find its isomorphisms.

(ii) Take as objects the finite-dimensional complex vector spaces, and as morphisms $V \to W$ the linear maps of rank at most 1, composed as maps. Show that a composite of two such maps is again one. Show that this is not a category, and find the objects V that have an identity morphism.

(iii) Show that a functor $BG \to BH$ is exactly a homomorphism $G \to H$. How many functors are there $B\mathbb{Z}_4 \to B\mathbb{Z}_2$, and how many $B\mathbb{Z}_2 \to B\mathbb{Z}_3$?

(iv) Let $\varphi, \psi \colon G \to H$ be homomorphisms, read as functors $BG \to BH$. Show that a natural transformation $\varphi \Rightarrow \psi$ is an element $h \in H$ with $\psi(g) = h\,\varphi(g)\,h^{-1}$ for all $g \in G$, and that it is always a natural isomorphism. There are 10 homomorphisms $S_3 \to S_3$; how many classes do they form up to natural isomorphism?

Solution. (i) Write $m \to n$ for the unique morphism when $m \leq n$. If $m \leq n$ and $n \leq p$ then $m \leq p$, so the composite exists and is the unique morphism $m \to p$; $m \leq m$ gives the identity. Every Hom set has at most one element, so any two morphisms with the same source and target are equal: associativity and the unit laws hold automatically. An isomorphism $m \to n$ needs a morphism $n \to m$ as well, so $m \leq n \leq m$ and $m = n$: only the identities are isomorphisms.

(ii) If $f \colon U \to V$ has rank at most 1, then $\mathrm{im}(g \circ f) = g(\mathrm{im}\, f)$ has dimension at most $\dim \mathrm{im}\, f \leq 1$, for every linear g. Composition of maps is associative. Now suppose $e \colon V \to V$ is an identity for V in this collection. For each nonzero $v \in V$ take the rank-one map $f \colon \mathbb{C} \to V$, $f(1) = v$; then $e \circ f = f$ gives $e(v) = v$. So $e = \mathrm{id}_V$, whose rank is $\dim V$: an identity exists exactly when $\dim V \leq 1$ (for $\dim V = 0, 1$, id_V itself has rank at most 1). Since $\mathbb{C}^2$ is an object with no identity, this is not a category.

(iii) As in the proof of Theorem 1.9, a functor $F \colon BG \to BH$ sends $*$ to $*$ and each g to $F(g) \in H$, with $F(gh) = F(g)F(h)$ and $F(1) = 1$: a homomorphism. Conversely a homomorphism φ satisfies both axioms ($\varphi(1) = 1$ follows from $\varphi(1) = \varphi(1)\varphi(1)$). A homomorphism from a cyclic group is fixed by the image x of a generator, and any x whose order divides the group's order is allowed. In $\mathbb{Z}_2$ both elements satisfy $4x = 0$, so there are 2 functors $B\mathbb{Z}_4 \to B\mathbb{Z}_2$; in $\mathbb{Z}_3$ only 0 satisfies $2y = 0$, so there is 1 functor $B\mathbb{Z}_2 \to B\mathbb{Z}_3$.

(iv) A natural transformation has one component $h = \xi_* \in H$, and naturality for the morphism g reads $\psi(g) \circ h = h \circ \varphi(g)$, that is $\psi(g) = h\varphi(g)h^{-1}$. Every element of H is invertible, so every natural transformation is a natural isomorphism, and two homomorphisms are naturally isomorphic exactly when they are conjugate by an element of H. For $S_3 \to S_3$, sort by the kernel, a normal subgroup: S_3, A_3 or $\{1\}$. Kernel S_3: the trivial homomorphism, 1 of

them. Kernel A_3: the image has order 2, generated by one of the three transpositions; the three are conjugate. Kernel $\{1\}$: the automorphisms of S_3. An automorphism permutes the three transpositions (the elements of order 2) and, since they generate S_3, is determined by that permutation; so there are at most six. The six conjugations by elements of S_3 are distinct (only 1 commutes with everything), so these are all, and each is conjugate to the identity. That is $1 + 3 + 6 = 10$ homomorphisms in 3 classes.

Exercise 2 (G-sets are functors $BG \to \mathrm{Set}$). A *G-set* is a set X with an action $g \cdot x$, such that $1 \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$; a map $f\colon X \to Y$ of G-sets is *equivariant* if $f(g \cdot x) = g \cdot f(x)$.

(i) Show that a functor $BG \to \mathrm{Set}$ is a G-set, and that a natural transformation between two such functors is an equivariant map.

(ii) Let $G = S_3$, let $X = \{1, 2, 3\}$ with the permutation action, and let $Y = \{+, -\}$ on which the even permutations act trivially and the odd ones swap $+$ and $-$. Count the equivariant maps $X \to X$, $X \to Y$, $Y \to X$ and $Y \to Y$.

(iii) For a finite G-set X, let $\mathbb{C}[X]$ have basis e_x ($x \in X$) and $\rho(g)e_x = e_{g\cdot x}$. Show that $\mathbb{C}[X]$ is a representation and that an equivariant $f\colon X \to Y$ gives an intertwiner $\mathbb{C}[f]\colon e_x \mapsto e_{f(x)}$, and that $X \mapsto \mathbb{C}[X]$, $f \mapsto \mathbb{C}[f]$ is a functor from finite G-sets to $\mathrm{Rep}(G)$.

Solution. (i) A functor $F\colon BG \to \mathrm{Set}$ is a set $X = F(*)$ and a map $F(g)\colon X \to X$ for each g, with $F(gh) = F(g) \circ F(h)$ and $F(1) = \mathrm{id}_X$. Setting $g \cdot x = F(g)(x)$, these two equations are exactly the two axioms of an action, and conversely an action defines such an F. A natural transformation $\xi\colon F \Rightarrow F'$ has the single component $f = \xi_*\colon X \to X'$, and naturality for g reads $F'(g) \circ f = f \circ F(g)$, that is $g \cdot f(x) = f(g \cdot x)$: equivariance. So G-sets with equivariant maps are $\mathrm{Fun}(BG, \mathrm{Set})$.

(ii) First a principle. Let X be one orbit, $x_0 \in X$, and $\mathrm{Stab}(x_0) = \{g : g \cdot x_0 = x_0\}$. An equivariant f satisfies $f(g \cdot x_0) = g \cdot f(x_0)$, so it is determined by $y_0 = f(x_0)$, and y_0 is fixed by $\mathrm{Stab}(x_0)$: $y_0 = f(s \cdot x_0) = s \cdot y_0$. Conversely, if y_0 is fixed by $\mathrm{Stab}(x_0)$, then $f(g \cdot x_0) = g \cdot y_0$ is well defined (if $g \cdot x_0 = g' \cdot x_0$ then $g^{-1}g'$ fixes x_0, hence y_0), and equivariant. So equivariant maps $X \to Z$ correspond to the points of Z fixed by $\mathrm{Stab}(x_0)$.

Both X and Y are single orbits. In X, $\mathrm{Stab}(1) = \{1, (2\,3)\}$. Its fixed points in X are $\{1\}$: one map $X \to X$, the identity. In Y the odd permutation $(2\,3)$ swaps $+$ and $-$, so it fixes no point: no map $X \to Y$. In Y, $\mathrm{Stab}(+) = A_3$, which fixes no point of X (a 3-cycle moves every point): no map $Y \to X$. And A_3 fixes both points of Y: two maps $Y \to Y$, the identity and the swap. The counts are $1, 0, 0, 2$ (of the 27, 8, 9 and 4 maps in all).

(iii) $\rho(g)\rho(h)e_x = e_{g\cdot(h\cdot x)} = e_{(gh)\cdot x} = \rho(gh)e_x$ and $\rho(1)e_x = e_x$, so ρ is a homomorphism into $\mathrm{GL}(\mathbb{C}[X])$. For equivariant f, $\mathbb{C}[f]\rho_X(g)e_x = e_{f(g\cdot x)} = e_{g\cdot f(x)} = \rho_Y(g)\mathbb{C}[f]e_x$, so $\mathbb{C}[f]$ is an intertwiner. Finally $\mathbb{C}[f' \circ f]e_x = e_{f'(f(x))} = \mathbb{C}[f']\mathbb{C}[f]e_x$ and $\mathbb{C}[\mathrm{id}_X] = \mathrm{id}$: a functor.

Exercise 3 (Restriction is a functor, faithful and not full). Let $H \leq G$ be a subgroup and $\iota\colon H \to G$ the inclusion.

(i) Show that restriction, $\mathrm{Res}(V, \rho) = (V, \rho \circ \iota)$ and $\mathrm{Res}(T) = T$, is a functor $\mathrm{Rep}(G) \to \mathrm{Rep}(H)$, and that on objects, read through Theorem 1.9, it is composition with the functor $B\iota\colon BH \to BG$.

(ii) Show that Res is faithful.

(iii) Let $G = S_3$, $H = A_3$, and $V = \{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ with the permutation action, the standard representation. Show that $\dim \mathrm{Hom}_{S_3}(V, V) = 1$ and $\dim \mathrm{Hom}_{A_3}(\mathrm{Res}\, V, \mathrm{Res}\, V) = 2$. Conclude that Res is not full.

(iv) Write down an intertwiner of $\mathrm{Res}\, V$ that does not come from $\mathrm{Hom}_{S_3}(V, V)$.

Solution. (i) If $T\rho_V(g) = \rho_W(g)T$ for all $g \in G$, then in particular for all $g \in H$; so Res sends morphisms to morphisms, and it preserves composites and identities because it does not change them. Reading (V, ρ) as a functor $F\colon BG \to \mathrm{Vect}_{\mathrm{fd}}$, the restricted representation is $F \circ B\iota$, where $B\iota$ sends $*$ to $*$ and h to $\iota(h)$ (a functor by Exercise 1(iii)).

(ii) $\mathrm{Hom}_G(V, W) \to \mathrm{Hom}_H(\mathrm{Res}\, V, \mathrm{Res}\, W)$ is the inclusion $T \mapsto T$, which is injective.

(iii) V is the standard two-dimensional irreducible representation of S_3, so by Schur's lemma its intertwiners are the multiples of id_V: dimension 1. For $A_3 = \{1, c, c^2\}$ with $c = (1\,2\,3)$: $\rho(c)^3 = \mathrm{id}$, and the trace of $\rho(c)$ on V is the trace of the permutation matrix on $\mathbb{C}^3$ minus the trace on the line of $(1, 1, 1)$, $0 - 1 = -1$. The two eigenvalues are cube roots of unity with sum -1, and the only such pair is ω, ω^2: they are distinct. A linear map commutes with all of A_3 exactly when it commutes with $\rho(c)$, which is diagonalisable with distinct eigenvalues; those maps are the ones diagonal in its eigenbasis: dimension 2. Since $2 > 1$, the map $\mathrm{Hom}_{S_3}(V, V) \to \mathrm{Hom}_{A_3}(\mathrm{Res}\, V, \mathrm{Res}\, V)$ is not surjective, and Res is not full.

(iv) $P_\omega = \frac{1}{3}(\mathrm{id} + \bar{\omega}\rho(c) + \omega\rho(c)^2)$ is a polynomial in $\rho(c)$, so it commutes with A_3. On an eigenvector with eigenvalue ω it gives $\frac{1}{3}(1 + \bar{\omega}\omega + \omega\omega^2) = 1$, and on one with eigenvalue ω^2 it gives $\frac{1}{3}(1 + \bar{\omega}\omega^2 + \omega\omega^4) = \frac{1}{3}(1 + \omega + \omega^2) = 0$. So P_ω is the projection onto the ω-eigenline, of rank 1: not a multiple of id_V, hence not an intertwiner for S_3.

Exercise 4 (The dual over the reals). Let $\mathrm{Vect}_{\mathbb{R}}^{\cong}$ be the groupoid of finite-dimensional real vector spaces and linear isomorphisms, with $D(V) = V^*$ and $D(f) = (f^*)^{-1}$ as in Lecture 1.

(i) Show that no family of linear isomorphisms $\theta_V\colon V \to V^*$ is a natural transformation $\mathrm{id} \Rightarrow D$ on $\mathrm{Vect}_{\mathbb{R}}^{\cong}$.

(ii) On the groupoid of finite-dimensional real inner-product spaces and orthogonal maps, show that $R_V(u) = \langle u, \cdot \rangle$ is a natural isomorphism $\mathrm{id} \Rightarrow D$. Why does this contradict neither (i) nor Theorem 1.11?

(iii) For which real numbers λ does the test $f = \lambda\, \mathrm{id}_V$ prove (i)? What plays the role of λ in the lecture's proof for unitary maps, and why is there no such scalar among the orthogonal maps?

Solution. (i) As in the lecture, naturality of θ for $f\colon V \to W$ reads $(f^*)^{-1} \circ \theta_V = \theta_W \circ f$, that is $f^* \circ \theta_W \circ f = \theta_V$. Take $V = W \neq 0$ and $f = 2\,\mathrm{id}_V$. Then $f^* = 2\,\mathrm{id}_{V^*}$ and, θ_V being linear, $f^* \circ \theta_V \circ f = 4\,\theta_V$. Naturality demands $4\theta_V = \theta_V$, so $\theta_V = 0$, which is not an isomorphism.

(ii) The real inner product is bilinear, so R_V is a linear map $V \to V^*$, and it is injective (if $\langle u, \cdot \rangle = 0$ then $\langle u, u \rangle = 0$, so $u = 0$), hence an isomorphism by dimension. For orthogonal $f\colon V \to W$ and $u, v \in V$,

$$f^*\big(R_W(fu)\big)(v) = R_W(fu)(fv) = \langle fu, fv \rangle = \langle u, v \rangle = R_V(u)(v),$$

so $f^* \circ R_W \circ f = R_V$: natural. There is no contradiction with (i), because this groupoid has fewer morphisms, and the test map $2\,\mathrm{id}$ is not orthogonal. Nor with Theorem 1.11: that theorem is about complex spaces, where the Riesz map is antilinear (Example 1.13) and so not a morphism of Vect; over $\mathbb{R}$ it is linear.

(iii) For $f = \lambda\,\mathrm{id}_V$ with $\lambda \neq 0$, $f^* \circ \theta_V \circ f = \lambda^2\theta_V$, and naturality forces $\theta_V = 0$ exactly when $\lambda^2 \neq 1$. For $\lambda = -1$ the test says nothing. The lecture's proof for unitary maps uses $\lambda = i$: unitary, with $\lambda^2 = -1 \neq 1$. A real scalar λ gives an orthogonal map only if $\lambda^2 = 1$, so the orthogonal maps contain no usable scalar, and indeed R is natural for them.

Exercise 5 (Skeletons). (i) Let FinBij be the groupoid of finite sets and bijections, and S the category with objects $0, 1, 2, \ldots$, $\mathrm{Hom}(n, n) = S_n$ and no morphisms $m \to n$ for $m \neq n$. Show that S is a skeleton of FinBij, and that the two are not isomorphic.

(ii) Let $\mathcal{G}$ be a groupoid in which any two objects are isomorphic, and x an object. Show that $\mathcal{G} \simeq B\,\mathrm{Aut}(x)$, where $\mathrm{Aut}(x) = \mathrm{Hom}(x,x)$. Apply this to the groupoid of the two-element subsets of $\{1,2,3\}$ with bijections.

(iii) In the proof of Theorem 1.16 the bases were chosen with $b_{\mathbb{C}^n} = \mathrm{id}$. Keep every choice but one: $b_{\mathbb{C}^2} = \mathrm{diag}(1,2)$. Compute $G(M(A))$ for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Is $G \circ M = \mathrm{id}$? Are M and G still an equivalence?

(iv) Lecture 1's functor "forgetting crossings" goes from the braid groupoid Br to the category S of (i). Show that it is full and not faithful. You may use one fact about braids: two braids on the same strands are equal in Br only if they have the same *exponent sum*, the number of positive crossings minus the number of negative ones, where a crossing is positive when the strand going up to the right passes over.

Solution. (i) Let $F\colon \mathrm{S} \to \mathrm{FinBij}$ send n to $\{1,\dots,n\}$ and a permutation to itself. It is a functor, and on Hom sets it is a bijection: $\mathrm{Hom}(n,n) = S_n$ is the set of bijections of $\{1,\dots,n\}$, and for $m \neq n$ both sides are empty. So F is full and faithful. Every finite set is in bijection with some $\{1,\dots,n\}$, so F is essentially surjective, and by Proposition 1.17 it is an equivalence. S is skeletal: isomorphic objects $m \cong n$ need a morphism $m \to n$, so $m = n$. Hence S is a skeleton of FinBij. They are not isomorphic: an isomorphism of categories is injective on objects and keeps isomorphic objects isomorphic (Lemma 1.6); $\{1\}$ and $\{2\}$ are distinct isomorphic objects of FinBij, while no two distinct objects of S are isomorphic.

(ii) Let $J\colon B\,\mathrm{Aut}(x) \to \mathcal{G}$ send $*$ to x and $a \in \mathrm{Aut}(x)$ to itself. On Hom sets it is the identity of $\mathrm{Hom}(x,x)$: full and faithful. Every object is isomorphic to $x = J(*)$: essentially surjective. By Proposition 1.17, J is an equivalence. Explicitly, choose $b_y\colon x \to y$ for each y, with $b_x = \mathrm{id}_x$, and set $K(g) = b_{y'}^{-1} \circ g \circ b_y$ for $g\colon y \to y'$, a functor $K\colon \mathcal{G} \to B\,\mathrm{Aut}(x)$; this is the proof of Theorem 1.16 again. The two-element subsets of $\{1,2,3\}$ are three objects, any two in bijection, and each Hom set has two bijections; $\mathrm{Aut}(\{1,2\}) \cong S_2$, so the groupoid is equivalent to BS_2, though not isomorphic to it (three objects against one).

(iii) $G(M(A)) = b_{\mathbb{C}^2}^{-1}\, A\, b_{\mathbb{C}^2}$ for a 2×2 matrix A, and

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \neq A.$$

So $G \circ M \neq \mathrm{id}$. The lecture's proof that G is a functor did not use $b_{\mathbb{C}^n} = \mathrm{id}$ (the b's cancel), and b is still a natural isomorphism $M \circ G \Rightarrow \mathrm{id}$. For the other composite, the matrices $\eta_n = b_{\mathbb{C}^n}^{-1}$ (read as morphisms $n \to n$ of $\mathrm{Mat}_{\mathbb{C}}$) form a natural isomorphism $\mathrm{id} \Rightarrow G \circ M$: for $A\colon m \to n$,

$$G(M(A))\,\eta_m = b_{\mathbb{C}^n}^{-1} A\, b_{\mathbb{C}^m} b_{\mathbb{C}^m}^{-1} = \eta_n\, A.$$

An equivalence asks only for natural isomorphisms, so M and G are still one; the choice $b_{\mathbb{C}^n} = \mathrm{id}$ only made $G \circ M$ equal to id on the nose.

(iv) Both categories have morphisms only from n to n, so fullness asks that every $\pi \in S_n$ be the permutation of some braid on n strands. The braid σ_i that crosses the strands at i and $i+1$ once and leaves the others straight forgets to the transposition $(i\ i{+}1)$. Every permutation is a product of such transpositions of neighbours (sort by swapping neighbours), and the functor sends a composite of braids to the composite of their permutations; so the corresponding product of σ_i's forgets to π. The functor is full. It is not faithful: on two strands, $\sigma_1 \circ \sigma_1$ and the identity braid both forget to the identity permutation, since the strand that starts at 1 ends at 1. Their exponent sums are ± 2 and 0, so by the fact quoted they are different morphisms of Br.

Exercise 6 ⋆. *Natural transformations of the identity of* $\mathrm{Rep}(G)$. Let G be finite, $\mathbb{C}[G]$ its group algebra (basis $g \in G$, product extended linearly), and ρ_V extended linearly to $\mathbb{C}[G]$. A natural transformation $\xi\colon \mathrm{id}_{\mathrm{Rep}(G)} \Rightarrow \mathrm{id}_{\mathrm{Rep}(G)}$ is an intertwiner $\xi_V\colon V \to V$ for each representation V, with $T \circ \xi_V = \xi_W \circ T$ for every intertwiner $T\colon V \to W$.

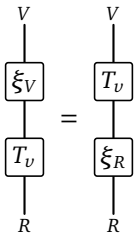
(i) For z in the centre $Z(\mathbb{C}[G])$, show that $\xi_V = \rho_V(z)$ is such a natural transformation.

(ii) Conversely, let ξ be one, let $R = \mathbb{C}[G]$ be the regular representation, $\rho_R(g)a = ga$, and put $z = \xi_R(1)$. Draw the naturality of ξ for an intertwiner $T\colon R \to V$ as a box sliding through ξ, in the style of Lecture 1's Equation (1.2). Show that z is central and that $\xi_V = \rho_V(z)$ for every V.

(iii) Show that $z = \sum_g z_g g$ is central exactly when z_g is constant on conjugacy classes. Natural transformations of $\mathrm{id}_{\mathrm{Rep}(G)}$ are added and scaled component by component, and composed as in Lemma 1.8. Conclude that they form an algebra isomorphic to $Z(\mathbb{C}[G])$, of dimension the number of conjugacy classes, and give that number for S_3 and for the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Solution. (i) ρ_V is a homomorphism of algebras $\mathbb{C}[G] \to \mathrm{End}(V)$, so $\rho_V(z)\rho_V(g) = \rho_V(zg) = \rho_V(gz) = \rho_V(g)\rho_V(z)$: $\rho_V(z)$ is an intertwiner. An intertwiner $T\colon V \to W$ satisfies $T\rho_V(g) = \rho_W(g)T$ for each g, hence, by linearity, $T\rho_V(a) = \rho_W(a)T$ for every $a \in \mathbb{C}[G]$; with $a = z$ this is naturality.

(ii) For $v \in V$, $T_v(a) = \rho_V(a)v$ is linear and $T_v(ga) = \rho_V(g)\rho_V(a)v = \rho_V(g)T_v(a)$, an intertwiner $R \to V$. Naturality, $\xi_V \circ T_v = T_v \circ \xi_R$, is the box T_v sliding through ξ, as in Equation (1.2):



At $a = 1$ it gives

$$\xi_V(v) = \xi_V(T_v(1)) = T_v(\xi_R(1)) = \rho_V(z)v.$$

With $V = R$ and $v = g$, this reads $\xi_R(g) = zg$. On the other hand ξ_R is an intertwiner, so $\xi_R(g) = \xi_R(\rho_R(g)1) = \rho_R(g)\xi_R(1) = gz$. Hence $gz = zg$ for every g, and by linearity z is central.

(iii) $hzh^{-1} = \sum_g z_g\, hgh^{-1} = \sum_g z_{h^{-1}gh}\, g$, so z is central, $hz = zh$ for all h, exactly when $z_{h^{-1}gh} = z_g$ for all g, h: constant on conjugacy classes. The class sums are therefore a basis of $Z(\mathbb{C}[G])$, which has dimension the number of classes. By (i) and (ii), $\xi \mapsto \xi_R(1)$ and $z \mapsto \rho(z)$ are inverse to each other ($\rho_R(z)1 = z$). The first is linear, since sums and multiples are taken component by component, and sends id to 1; it respects composition: $(\xi \circ \xi')_R(1) = \xi_R(\xi'_R(1)) = \rho_R(z)z' = zz'$. So the natural transformations form an algebra isomorphic to $Z(\mathbb{C}[G])$. S_3 has the classes $\{1\}$, the three transpositions and the two 3-cycles: 3. Q_8 has $\{1\}$, $\{-1\}$, $\{\pm i\}$, $\{\pm j\}$, $\{\pm k\}$: 5.